

Problem 1

What do the results on the functioning of central affective mechanisms imply for the von Neumann-Morgenstern theory of risk behavior?

vNM expected utility theory treats risk attitude as a fixed property of a single, context-free utility function over wealth — the same curvature governs every choice regardless of emotional state. Neuroscience findings on affective mechanisms complicate this. Patients with damage to affect-generating regions (ventromedial prefrontal cortex, amygdala) make worse decisions under risk, not more “rational” ones (Damasio’s somatic marker hypothesis; the Iowa Gambling Task studies of Bechara et al.). Emotion is a necessary input to risk evaluation, not noise to be filtered out. This implies risk attitude is state-dependent — it shifts with arousal, mood, and framing — rather than a stable parameter, and that choices reflect an interaction between fast affective and slow deliberative systems (Kahneman’s dual-process view) rather than one coherent utility function. The independence axiom in particular breaks down once affect enters: identical gambles framed differently produce different choices. vNM theory may still hold as a normative ideal for a purely rational agent, but as a descriptive account of actual behavior it needs supplementing by models like prospect theory that build affect and framing into the decision process.

Problem 2

What role does the hormone or neuropeptide oxytocin play in cooperative behavior?

Oxytocin is a neuropeptide linked to social bonding that also shapes cooperative economic behavior. Kosfeld et al. (2005) found that subjects given intranasal oxytocin transferred more money to a trustee in a trust game than those given a placebo, while oxytocin did not increase risk-taking in a non-social gamble — its effect is specific to social trust, not general risk tolerance. The likely mechanism is that oxytocin dampens amygdala-driven fear of betrayal, lowering the psychological hurdle to cooperating. However, De Dreu et al. (2010) show this is not universal prosociality: oxytocin increases cooperation and generosity toward in-group members while increasing defensiveness and aggression toward out-groups (“parochial altruism”). For economics, this means cooperation in games like the prisoner’s dilemma or public-goods game depends not just on incentives (reputation, punishment, repeated play) but on an endogenous, biologically mediated trust mechanism that operates along in-group/out-group lines.

Problem 3

Assume that psychological and neuroscientific theories lend support to a virtue ethical conception of the good life. What does this imply for classical economic preference theory, and what implications can you derive for normative economics?

Classical preference theory assumes individuals hold a fixed, exogenous preference ordering over outcomes, and that welfare is simply preference satisfaction as revealed by choice. A virtue-ethical (Aristotelian) conception of the good life instead locates well-being in the exercise of virtuous character — practical wisdom, temperance, courage — cultivated through habituation, not in getting whatever one happens to want. If psychology and neuroscience support this view (habit formation, akrasia, dual-process reasoning, the affective and social embeddedness of choice discussed above), classical preference theory is undermined: preferences turn out to be endogenous and shaped by habit and character rather than fixed data, and revealed preference breaks down as a welfare measure because people routinely choose against their own considered well-being (addiction, present bias). For normative economics, this weakens the case for treating preference satisfaction and consumer sovereignty as the

ultimate welfare standard. It lends support to alternative frameworks that judge outcomes by objective flourishing or capability — e.g., Sen and Nussbaum's capability approach — and gives an economic rationale for policy that shapes choice architecture to support virtuous, self-consistent decisions (Thaler and Sunstein's "nudging"), rather than simply deferring to whatever preferences people reveal.

References

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